

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/363129982>

Drug–target Interaction Prediction Using Machine Learning

Conference Paper · December 2021

DOI: 10.1109/CICIS52592.2021.9694127

CITATIONS

5

READS

38

3 authors, including:



Mohamed Barkat

Ain Shams University

2 PUBLICATIONS 5 CITATIONS

[SEE PROFILE](#)



Sherin Moussa

Université Française d'Égypte - Ain Shams University

66 PUBLICATIONS 568 CITATIONS

[SEE PROFILE](#)

Drug-target Interaction Prediction Using Machine Learning

Mohamed R. Barkat

Bioinformatics Program
Department of Information Systems
Faculty of Computer and Information
Sciences
Ain Shams University
Cairo, Egypt
mohamed_barkat@cis.asu.edu.eg

Sherin M. Moussa

Department of Information Systems
Faculty of Computer and Information
Sciences
Ain Shams University
Cairo, Egypt
sherinmoussa@cis.asu.edu.eg

Nagwa L. Badr

Department of Information Systems
Faculty of Computer and Information
Sciences
Ain Shams University
Cairo, Egypt
nagwabadr@cis.asu.edu.eg

Abstract—In the last decade, there has been a major worldwide evolution in the drug discovery and design technologies, in conjunction with the bold emergence of bioinformatics. Different artificial intelligence and machine learning (ML) approaches have been adopted in drug discovery and design, causing a mutation in all their stages, such as drugs and compounds discrimination, compounds classification, and drug-target interaction (DTI) prediction, which is very time consuming. Researchers using positive and negative interactions for DTI prediction usually randomize negative data, considering unknown interaction as negative interactions. This may reduce the accuracy of DTI prediction. In this paper, a new method for drug-target interaction prediction is proposed using improved ML and feature extraction methods for faster and more accurate results. The negative dataset for drugs is generated based on the similarity among drugs, solving the problem of noisy negative data in the literature. The proposed method uses fewer features for drugs and protein targets, which makes DTI prediction and negative data generation faster. The experiments show that the DTIs prediction using the proposed method achieves an average accuracy of 95%, which outperforms the benchmark studies with an average of 3-4 %.

Keywords—drug discovery, drug design, machine learning, bioinformatics, Support Vector Machine, Drug-target interaction, Tree Model

I. INTRODUCTION

Drugs development is a time-consuming and expensive investment, in which a new drug consumes up to 15 years and more than \$800 million to develop [1]. Drugs work by altering the downstream processes of their targets. Classic drug discovery methodologies examine if a single protein is a pharmacological target for a given therapeutic indication using the paradigm “one molecule, one target, one disease” [2].

Drug targets are biological proteins from 130 protein families, with four categories of targets receiving the most attention: enzymes, ion channels, nuclear hormone receptors and G-protein-coupled receptors (GPCRs) [2-4]. The number of pharmacological targets in the human genome is believed to be between 6,000 and 8,000. As a result, detecting drug-target interactions (DTIs) has become a prerequisite in many fields, such as drug repositioning, drug resistance, drug development, side-effect prediction and polypharmacology [3], [5].

The automation of examining highly dimensional and noisy biological data has exceedingly promoted machine learning (ML) approaches in drug development [6],[7],[26]. Various ML tasks have been investigated, such as supervised learning, aiming to predict the label of new observations based on a vast database of previously labelled cases. Supervised learning techniques can be classified as (i) feature-based methods, such as chemical and genomic features [8], functional groups and biological features [9], or (ii) similarity-based methods. Similarity-based methods include matrix factorization [10], [11] and graph-based [12], [13]. A Support Vector Machine (SVM) [14] is one of the main supervised learning algorithms used in the biological environment [7]. An application of SVMs in drug discovery was presented for pharmaceutical data analysis [15], where the performance of SVM classification in the recovery of dihydrofolate reductase inhibitors with a large number of base learners, such ML models become time consuming and inefficient for prediction tasks [13], [27]. Random sampling for negative instances also causes many problems with respect to the selection accuracy of negative samples, but it is used because of the large size of data [13].

In this paper, we propose a framework for drug-target interaction prediction with reduced number of features, which reduces data size by removing noisy, or irrelevant information. We generate a negative dataset for drug interactions without depending on random selection for more accurate results. Thus, the main contributions of this study is summarized as follows:

- (1) Generate a learning model that uses the available DTIs for a set of drugs and proteins to predict drug interactions with other proteins. It can be formalized as:

Given: a set of proteins $P (P_1, P_2, \dots, P_n)$, a set of drugs $D (D_1, D_2, \dots, D_n)$, a set of positive interactions between members of P and D , and a set of proteins $D_2 (D_n + 1)$.

Find: all similar interactions between members of P and $D_2 (D_n + 1)$.

- (2) Generate a negative dataset for drugs from a positive dataset (negative sampling), instead of random selection, and without using alignment, formalized as:

Given: a set of drugs D , a set of positive interactions between members of D and D .

Generate: negative interactions between D and D members.

- (3) Reduced number of extracted features for DTIs that reduces the data size and consumed time.

The rest of the paper is organized as follows. Section II overviews the main related studies, highlighting their drawbacks and limitations. Section III presents our proposed method, with a detailed description for the solution provided to handle the deduced research gap. Section IV discusses the conducted experiments and the obtained results, whereas section V concludes our study and suggests some future directions to improve the accuracy of DTI prediction.

II. RELATED WORKS

Different feature-based or similarity-based research has developed several methods for DTIs prediction with various representations of input data in order to improve prediction accuracy. In feature-based ML, the input data are in the form of a vector of features, which is an DTI pair with its corresponding class label that indicates whether an interaction exists [13]. Examples include DTIs prediction from efficiently integrated chemical, genetic and pharmacological information based on Random Forest (RF) and Support Vector Machine (SVM) [8]. In [9], DTIs were predicted based on functional groups and biological features. Functional groupings encode drug molecules, while the biological aspects such as biochemical and physicochemical qualities encode proteins. Using ensemble learning in [16], and ensemble learning strategy was suggested that integrates techniques to handle new medicines and targets for newly unidentified interactions.

In such feature-based studies, DTI prediction does not consume much time with small datasets, but they face some drawbacks, i.e. positive examples were represented by known interactions, whereas negative instances were represented by non-interactions [13]. Negative occurrences in this context include both non-interactions and unknown DTIs (false negatives), which is inaccurate, since an unknown interaction may be a positive interaction [13]. Due to the large number of possible negative occurrences, feature-based approaches typically under-sample the negative data prior to training and prediction to balance the positive and negative training data, that is, negative instances are randomly sampled to have the same size as positive examples [13], [27].

On the other hand, similarity-based approaches accept their inputs directly in the form of two similarity matrices for drugs and targets, as well as an interaction matrix that shows which pairs of drugs and targets interact [13]. For instance, researchers in [11] presented a Bayesian framework to predict DTI networks based on matrix factorization using chemical similarity between drug compounds and genomic similarity between target proteins. This includes dimensionality reduction, matrix factorization, and binary classification. A Network-based Random Walk with Restart on the Heterogeneous Network (NRWRH) was presented in [12] to anticipate possible DTIs on a wide scale based on the

assumptions that similar drugs often target similar target. Prediction of DTI based on structure may be more accurate, but the difficulty of identifying and accessing the structure of proteins was considered as its main drawback [13].

Thus, in this study, the feature-based approaches are adopted due to the availability of data. The problem of noisy random selection of negative data is overcome by proposing a new method for negative data generation based on similarity between drugs. In addition, the longtime of DTI prediction is handled for huge datasets by considering fewer well-selected features.

III. METHODS

In this section, we propose a novel negative sampling method to generate a negative dataset for drugs. It depends on measuring the similarity between two vectors of extracted features, where each vector represents a drug. ML models learn from drug-protein (target) interacting pairs based on extracted features from each part. The features extraction methods for both interaction partners extract small number of features (7 for drug, 21 for protein), which reduces the time of the feature extraction process.

A. Feature Extraction

Features were extracted for both DTI partners, firstly for the target, that is, protein. We employed a protein feature extraction method based on the extraction of epitopes from protein sequences [17]. An antibody is a protein produced by the immune system that attaches to an antigen with a great specificity and then exclusively to a region of the protein called an epitope [17]. To reflect the structural and physico-chemical properties of peptides, sequences are first transformed into fixed length feature vectors. It divides the 20 amino acids into three categories depending on their hydrophobicity, resulting in three descriptors that reflect the epitopes' overall makeup as follows:

- (1) Composition (C), a vector made up of three real values, each of which represents the fraction of amino acids in each of the three groups,
- (2) Transition (T), a set of three real numbers that represent the frequency of amino acids from one group, followed and preceded by amino acids from another group,
- (3) Distribution (D), a vector made up of 15 real numbers, where the first, 25 percent, 50 percent, 75 percent and 100 percent of amino acids for each group in the sequence are represented by the first, 25%, 50%, 75%, and 100% of real numbers [17].

Each protein is then given its own set of 21 feature vectors. Secondly, features were extracted for drugs. a downloaded drug features [30] were used, in which mapping was applied on it. Each feature was divided into classes and each class was indicated with a numerical value. Each feature in the drug features was mapped to a numerical value, resulting that each drug in the dataset is replaced by a vector of numerical values expressing the drug features [30].

B. Generation of drug negative dataset using Dissimilarity

Research sampling. Our proposed method for negative generation aims to generate negative interactions that would decrease the expected false negative samples compared to random sampling. According to our hypotheses, if two drugs have dissimilar features, a protein that interacts with one of them can be considered as a negative sample for the other. Thus, we proposed a negative-random generation algorithm, which calculates the Manhattan distance among all vectors of drugs features. The distances are then normalized and considered as dissimilarity distances matrix for all drugs. The drugs having the lowest positive interactions with proteins are selected (idle drugs). Based on the application of dissimilarity average threshold and dissimilarity distances matrix, major false negative interactions are excluded. random sampling is then applied over the remaining negative interactions as shown in Algorithm1.

Dissimilarity threshold (D). Negative interactions are generated based on the idle drugs having the highest negative interactions. In our method, we find drugs having similarity with idle drugs greater than (D), which is the average of dissimilarity distances among all vectors of drugs features considering their interactions as negative interactions for idle drugs. The similarity is calculated based on the features of drugs, allowing the learning model for DTI prediction to be constructed.

C. Classification Model

SVMs are used for their classification power and ability to stand high noise for modelling nonlinearity connections [18],[19]. While the logit approach uses a maximum likelihood algorithm, SVMs aim to project input vectors on a binary (i.e., two classes) hyperplane that is linearly separable [22],[23]. SVMs can also discover a separation boundary for a classification by a curved (nonlinear) separation if a linear separation is not achievable. By increasing the dimensionality of the input variable space in SVMs, nonlinear solutions can be obtained [20]. We used kernel functions such as [28]:

Linear Kernel:

$$F(x, x_j) = \text{sum}(x, x_j) \quad (1)$$

where x, x_j reflects the data needed to classify.

Polynomial Kernel:

$$F(x, x_j) = (x \cdot x_j + 1)^d \quad (2)$$

where ' $\cdot$ ' is the dot product of both values, d is the degree, $F(x, x_j)$ is the decision boundary to separate the given classes.

RBf Kernel:

$$F(x, x_j) = \exp(-\text{gamma} * ||x - x_j||^2) \quad (3)$$

The gamma parameter determines how far a single training example's influence extends, with low values indicating "far" and large values indicating "near" [28]. The gamma value ranges from 0 to 1. The gamma value of 0.1 is the most preferred [28]. In addition, Linear model [21] and decision tree

model [22] are used.

Algorithm1: negative-random generation for drugs

Input: Negative drugs vector (contains drugs having the lowest positive interactions in positive interaction database), distances matrix (contains distances between the idle drugs and all proteins.), PositiveSet (dictionary that contains all drugs and their interactions)

Output: Negative interactions dictionary

Initialize NegativeSet as empty dictionary for saving negative interactions

```

1. Begin
2.   Initialize Drugs list as empty list // contains header row of distances
3.   For each I in Idle drugs // loop for all Idle drugs
4.     Initialize Dsdrugs as empty list //to save all proteins that have
       dissimilarity distances with I more than Dissimilarity average threshold (D).
5.     For each J in Drugs_List
6.       If (dissimilarity distance between I and J
7.         > (D) = True) Then add J for Dsdrugs //get dissimilarity distance
       from distances and check if distance is more than D.
8.     End if
9.   End for
10.  Initialize DsOFF as empty list // all proteins that make interaction with drugs
       in Dsdrugs
11.  For each S in Dsdrugs
12.    Initialize Interactions as empty list //proteins with interaction of drug S
13.    Interactions = PositiveSet [K].
14.    Add Interactions to DsOFF.
15.  End for
16.  Initialize Number count = length of PositiveSet [I].
17.  Initialize NegC = count.
18.  Initialize a list NegSet_D as empty list // to save negative interactions of the
       Idle Drug
19.  If (NegC < length (DsOFF) and NegC > 0) Then
20.    NegSet_D = random number of DsOFF equal NegC
21.  Else if (NegC <= length (DsOFF)) Then
22.    Set NegSet_D = DsOFF
23.  Else
24.    NegSet_D = empty list
25.  End if
26.  NegativeSet (I) = NegSet_D
27. End for
28.  Return NegativeSet
29. END

```

IV. EXPERIMENTAL EVALUATION

A. Experimental Settings and Methodology

Five machine learning techniques were applied on positive and negative datasets to build a learning model for DTI prediction as follows: SVM with RBF kernel function (RPF), Linear SVM (SVM), SVM with polynomial kernel function (SVM(P)), Tree Model (TM) and Linear Model (LM). Experiments were held on a machine with processor Intel(R) Core(TM) i7-7500U CPU 2.70GHz, 2.90 GHz and RAM 8.00 GB. In our experiments, the scikit-learn library ver. 0.24 was imported for machine learning support in Python programming language version 3.5.10.

B. Dataset

Drugbank was used as the drug target interaction dataset [30]. It consists of 1,264 interactions among 504 drugs and 507 proteins. The dataset includes drugs names, their accession numbers, proteins names, their UniprotId on Uniprot [29]. Drug features were obtained to get proteins sequences or extraction of protein features [30]. The generated negative dataset based on our proposed method is available at [31].

Drug features. Small molecule drugs are used, having chemically synthesized pharmaceuticals with a molecular weight of less than 900 Daltons. The structural and chemical characteristics of these medications are well defined. Some medications with a molecular weight more than 900 Daltons are classified as small molecule pharmaceuticals in Drugbank [30], such as deoxyribonucleotides, monosaccharides, and amino acids. Drugs are classified into groups, which specify where they are in the development process as follows [30]:

1. *Approved*: At some instance in time, a chemical drug was approved in at least one jurisdiction. This does not imply that the drug is now approved or available; rather, it indicates that it has been marketed and approved at some instance.
2. *Vet Approved*: A chemical drug that has been licensed for the treatment of animals in at least one country at some instance of time.
3. *Nutraceutical*: A pharmaceutical-grade, standardized nutrition drug, which has confirmed or unconfirmed health benefits.
4. *Illicit*: At some instance of time, a chemical drug that is scheduled in at least one jurisdiction.
5. *Withdrawn*: At some instance of time, a previously approved medicine was pulled off the market in at least one jurisdiction. It is important to note that a drug may be approved in one jurisdiction but withdrawn in another, so it is possible to be included in both drug groups (approved, withdrawn). If a drug was approved at some instance of time and then withdrawn, it would be included in both drug groups.
6. *Investigational*: This means a chemical drug is in some stage of the approval process in at least one jurisdiction.
7. *Experimental*: A chemical compound that has been demonstrated to bind specific proteins in mammals, fungi, viruses, parasites or bacteria in an experimental setting.
8. *Solubility state of a compound*: The propensity of a molecule to behave as a solute in a liquid medium to generate a solution in which the solute molecules are scattered and individually surrounded by solvent molecules. The solute can come from a gaseous, liquid, or solid form [23].

In addition, we used other taxonomy of drugs as features, such as Kingdom, SuperClass, Class, and SubClass as a representation for the first, second, third and fourth levels of the chemical taxonomy of drugs, respectively. At the highest level (Kingdom), chemicals are divided into two categories: organic compounds and inorganic compounds. Organic compounds are chemical compounds with one or more carbon atoms in their structure. Inorganic compounds are defined as chemical compounds that are not organic, with the exception of a small number of "special" chemical

compounds, such as cyanide/isocyanide and their respective carbon dioxide, non-hydrocarbyl derivatives, carbon sulfide, carbon disulfide and carbon monoxide [24], [25].

On the other hand, SuperClasses (26 organic and 5 inorganic categories are included) consist of generic groups of chemical compounds with common structural identities (e.g., organometallic compounds, organic acids and derivatives, lipids and lipid-like molecules, organic oxygen compounds, homogeneous metal compounds, nucleosides, nucleotides, and analogues). Each of which covers millions of chemicals that have been identified.

The next level below the SuperClass is the Class level, which includes 764 nodes. Classes are divided into specific chemical categories with more definite and recognized structural properties as actinide salts, pyrimidine nucleosides, thiadiazines, flavanols, benzazepines, oxazinanes. There are typically over 100,000 identified compounds in chemical classes. The next level below Classes represents SubClasses, containing more over 10,000 identified chemicals. In the present taxonomy, there are 1,729 SubClasses, having further 2,296 more categories that encompass taxonomic levels 5–11 [24],[30]. The direct parent is the category that is determined by the compound's greatest structural feature. For example, Imovane with Generic Name in drug bank Zopiclone has its direct parent Cyclopyrrolones [24].

C. Experimental Results and Discussion

The machine learning methods, RPF, SVM, SVM(P), TM and LM, were applied on the positive and generated negative interaction datasets after converting them to numerical vectors composed of two partitions: protein partition (21 features) and drug partition (7 features). Each DTI represented by 28 features to test the results using the generated negative dataset.

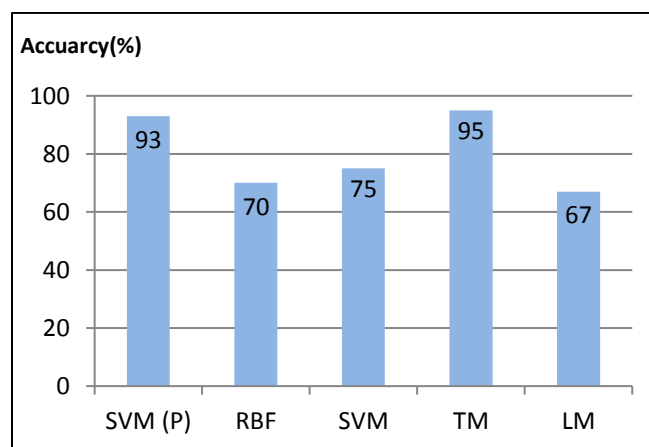


Fig. 1: Results of drug target interaction (DTI) prediction

The confusion matrix was used to calculate the overall rate of predictions that were correct as the accuracy metric. As presented in Fig. 1, TM has the best accuracy, that is 3% higher than SVM(P), which consumed an extra average of 15 minutes. This experiment indicates that TM is in line with the data nature more than the rest of ML methods, achieving 95%

as the best accuracy. It also took less time compared to the rest of methods. It can also be deduced from our experiment that the data nature is overlapping, as the LM was the least accurate method.

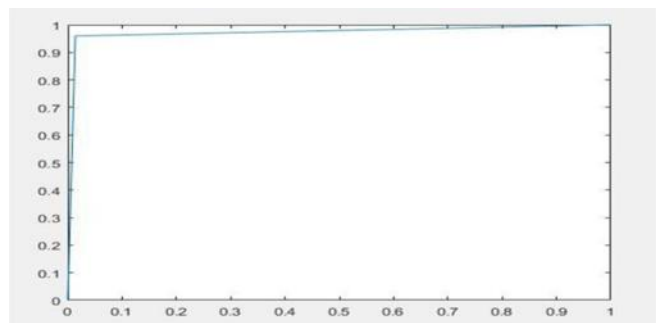


Fig. 2: AUC of the DTIs generated model

As shown in Fig. 2, since more verifications are important, the Receiver Operating Characteristic (ROC) curve was plotted by importing GridSearchCv library in python version 3.5.10. to calculate the Area Under the Curve (AUC) for the generated model and its value was 97.29%. Hence, the proposed method achieves highly accurate results while relying on the most effective features that helps distinguish between negative and positive interactions. Using the proposed method to generate negative data based on the similarity among drugs helps reducing the rate of false negative data, which improved the accuracy of DTI prediction.

V. CONCLUSION

This paper proposes a new method for drug-target interaction (DTI) prediction using improved ML approaches and feature extraction methods for faster and more accurate results with less features. The proposed method uses only 7 features for drugs and 21 features for protein targets, allowing DTI prediction and negative data generation to be a faster process. The proposed method generates the negative dataset for drugs based on the similarity among drugs, which reduces the noise in the generated negative data compared to the random approaches as in the literature for more accurate results. The experiments show that DTIs prediction using the proposed method achieves an average accuracy of 95%, which outperforms the benchmark studies with an average of 3-4 %. This indicates that it can be counted on the extracted features from protein sequences and classification features of drugs, as well as our proposed method for generating negative data to predict DTIs.

As our future work, we plan to examine new features of protein, such as its structural features, which could improve the accuracy of prediction. We also plan to depend on proteins to generate the negative dataset.

REFERENCES

- [1] C. P. Adams and V. van Brantner, "Market watch : Estimating the cost of new drug development: Is it really \$802 million?," *Health Affairs*, vol. 25, no. 2, pp. 420–428, Mar. 2006. doi: 10.1377/hlthaff.25.2.420.
- [2] W. Zhang, W. Lin, D. Zhang, S. Wang, J. Shi, and Y. Niu, "Recent Advances in the Machine Learning-Based Drug-Target Interaction Prediction," *Current Drug Metabolism*, vol. 20, no. 3, pp. 194–202, Aug. 2018, doi: 10.2174/1389200219666180821094047.
- [3] R. Chen, X. Liu, S. Jin, J. Lin, and J. Liu, "Machine learning for drug-target interaction prediction," *Molecules*, vol. 23, no. 9. MDPI AG, Aug. 31, 2018. doi: 10.3390/molecules23092208.
- [4] Klabunde T (2006) Chemogenomics approaches to ligand design. *LigandDesign for G Protein-coupled Receptors* ch7: 115–135..
- [5] A. Masoudi-Nejad, Z. Mousavian, and J. H. Bozorgmehr, "Drug-target and disease networks: polypharmacology in the post-genomic era," 2013. [Online]. Available: <http://www.in-silico-pharmacology.com/content/1/1/17>
- [6] S. Ekins *et al.*, "Exploiting machine learning for end-to-end drug discovery and development," *Nature Materials*, vol. 18, no. 5. Nature Publishing Group, pp. 435–441, May 01, 2019. doi: 10.1038/s41563-019-0338-z.
- [7] C. Réda, E. Kaufmann, and A. Delahaye-Duriez, "Machine learning applications in drug development," *Computational and Structural Biotechnology Journal*, vol. 18. Elsevier B.V., pp. 241–252, Jan. 01, 2020. doi: 10.1016/j.csbj.2019.12.006.
- [8] H. Yu *et al.*, "A systematic prediction of multiple drug-target interactions from chemical, genomic, and pharmacological data," *PLoS ONE*, vol. 7, no. 5, May 2012, doi: 10.1371/journal.pone.0037608.
- [9] Z. He *et al.*, "Predicting drug-target interaction networks based on functional groups and biological features," *PLoS ONE*, vol. 5, no. 3, Mar. 2010, doi: 10.1371/journal.pone.0009603.
- [10] Association for Computing Machinery. Special Interest Group on Management of Data and Association for Computing Machinery. Special Interest Group on Knowledge Discovery & Data Mining, *KDD '13 : the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining : August 11-14, 2013, Chicago, Illinois, USA*.
- [11] M. Gönen, "Predicting drug-target interactions from chemical and genomic kernels using Bayesian matrix factorization," *Bioinformatics*, vol. 28, no. 18, pp. 2304–2310, Sep. 2012, doi: 10.1093/bioinformatics/bts360.
- [12] X. Chen, M. X. Liu, and G. Y. Yan, "Drug-target interaction prediction by random walk on the heterogeneous network," *Molecular BioSystems*, vol. 8, no. 7, pp. 1970–1978, Jul. 2012, doi: 10.1039/c2mb00002d.
- [13] A. Ezzat, M. Wu, X. L. Li, and C. K. Kwoh, "Drug-target interaction prediction using ensemble learning and dimensionality reduction," *Methods*, vol. 129, pp. 81–88, Oct. 2017, doi: 10.1016/j.ymeth.2017.05.016.
- [14] W. S. Noble, "Support vector machine applications in computational biology," [Online]. Available: www.gs.washington.edu/noble/name-change. [Last accessed: 10-SEP-2021].
- [15] R. Burbidge, M. Trotter, B. Buxton, and S. Holden, "Drug design by machine learning: support vector machines for pharmaceutical data analysis," 2001. [Online]. Available: www.elsevier.com/locate/compchem [Last accessed: 15-MAY-2021].
- [16] A. Ezzat, M. Wu, X. L. Li, and C. K. Kwoh, "Drug-target interaction prediction via class imbalance-aware ensemble learning," *BMC Bioinformatics*, vol. 17, Dec. 2016, doi: 10.1186/s12859-016-1377-y.
- [17] J. Cui *et al.*, "Prediction of MHC-binding peptides of flexible lengths from sequence-derived structural and physicochemical properties," *Molecular Immunology*, vol. 44, no. 5, pp. 866–877, 2007. Available: 10.1016/j.molimm.2006.04.001.
- [18] B. Schölkopf, C. Burges and V. Vapnik, "Incorporating invariances in support vector learning machines," *Artificial Neural Networks — ICANN 96*, pp. 47–52, 1996. doi: 10.1007/3-540-61510-5_12.
- [19] *2011 International Conference on Information Fusion*. IEEE, 2011.
- [20] T. Verplancke *et al.*, "Support vector machine versus logistic regression modeling for prediction of hospital mortality in critically ill patients with haematological malignancies," *BMC Medical Informatics and Decision Making*, vol. 8, 2008, doi: 10.1186/1472-6947-8-56.
- [21] S. Liu, C. Liu, and L. Deng, "Machine learning approaches for protein-protein interaction hot spot prediction: Progress and comparative assessment," *Molecules*, vol. 23, no. 10, 2018, doi: 10.3390/molecules23102535.
- [22] N. Patel and D. Singh, "An Algorithm to Construct Decision Tree for Machine Learning based on Similarity Factor," *International Journal of Computer Applications*, vol. 111, no. 10, pp. 22–26, 2015, doi: 10.5120/19575-1376.
- [23] R. J. Young, "Physical properties in drug design," *Topics in Medicinal Chemistry*, vol. 9, pp. 1–68, 2014, doi: 10.1007/7355_2013_35.
- [24] Y. Djoumbou Feunang *et al.*, "ClassyFire: automated chemical classification with a comprehensive, computable taxonomy," *Journal of Cheminformatics*,

- vol. 8, no. 1, pp. 1–20, Nov. 2016, doi: 10.1186/s13321-016-0174-y.
- [25] F. Mao *et al.*, “Chemical structure-related drug-like criteria of global approved drugs,” *Molecules*, vol. 21, no. 1, Jan. 2016, doi: 10.3390/molecules21010075.
- [26] S. Redkar, S. Mondal, A. Joseph and K. Hareesha, "A Machine Learning Approach for Drug-target Interaction Prediction using Wrapper Feature Selection and Class Balancing", *Molecular Informatics*, vol. 39, no. 5, p. 1900062, 2020. doi: 10.1002/minf.201900062 [Last accessed 30-JUL-2021].
- [27] L. Xu, X. Ru, and R. Song, “Application of machine learning for drug-target interaction prediction,” *Front. Genet.*, vol. 12, p. 680117, 2021.
- [28] Dataaspirant, <https://dataaspirant.com/svm-kernels>. [last accessed: 3-JUL-2021]
- [29] Uniprot, <https://www.uniprot.org/>. [Last accessed: 3-SEP-2021].
- [30] Drugbank <https://go.drugbank.com/>. [Last accessed: 18-SEP-2021].
- [31] M. Sayed Ramadan, Sherin Moussa, October 31, 2021, "Drug Target Negative Interactions Dataset", IEEE Dataport, doi: <https://dx.doi.org/10.21227/4319-ky48>.